Question	Answer	Mark	Partial Marks
18(b)	4.8 to 5.8	**2**	dep on a close attempt at a tangent **M1** for $\frac{\text{rise}}{\text{run}}$ also dep on close attempt at tangent
19(a)	12	**3**	**M1** for $y = k(x-1)^2$ oe **M1** for $y = \textit{their } k(7-1)^2$ oe
19(b)	divided by 3 oe	**1**	
20	3.9	**3**	**M2** for $5.2 \times \sqrt[3]{\frac{33.75}{80}}$ oe or **M1** for $\frac{\sqrt[3]{33.75}}{\sqrt[3]{80}}$ oe or $\frac{\sqrt[3]{80}}{\sqrt[3]{33.75}}$ oe or $\frac{h^3}{5.2^3} = \frac{33.75}{80}$ oe
21	$4x^2 + 3x - 85[=0]$ or $16y^2 - 113y + 7[=0]$ oe simplified	**M2**	**M1** for $4(x^2 - 18) + 3x = 13$ or $x^2 - 18 = \frac{13-3x}{4}$ or $y = \left(\frac{13-4y}{3}\right)^2 - 18$ oe or better
	correct method to solve *their* quadratic equation e.g. factors, quadratic formula, completing the square	**M1**	$\frac{-3 \pm \sqrt{3^2 - 4 \times 4 \times -85}}{2 \times 4}$ oe, $(4x - 17)(x + 5)$ $\frac{-(-113) \pm \sqrt{(-113)^2 - 4 \times 16 \times 7}}{2 \times 16}$ oe, $(16y - 1)(y - 7)$
	$x = -5$ $y = 7$ $x = \frac{17}{4}$ oe $y = \frac{1}{16}$ oe	**B2**	**B1** for one correct pair or two correct x values or two correct y values If B0 scored and at least 2 method marks scored, **SC1** for correct substitution of both of their x values or their y values into $4y + 3x = 13$ or $y = x^2 - 18$
22(a)(i)	cubic	**1**	
22(a)(ii)	reciprocal	**1**	

Question	Answer	Mark	Partial Marks
22(b)(i)	correct sine curve sketch through (0, 0), (180, 0) and (360, 0)	**2**	**M1** for correct sine curve shape through the origin
22(b)(ii)	203.6 and 336.4	**3**	**B2** for one correct or **M1** for $\sin x = -0.4$ oe If 0 or M1 scored, **SC1** for two reflex angles with a sum of 540 or two non-reflex angles with a sum of 180
23(a)	15	**1**	
23(b)	$\frac{1}{2}$ oe nfww	**2**	**M1** for $\frac{2+3}{2+1+3+4}$ oe or $1-\frac{4+1}{2+1+3+4}$ oe with either the numerator or denominator correct
24(a)	$\frac{1}{2}\mathbf{b}-\frac{2}{3}\mathbf{a}$	**2**	**B**1 for answer $\frac{1}{2}\mathbf{b}+k\mathbf{a}$ or $j\mathbf{b}-\frac{2}{3}\mathbf{a}$ or correct unsimplified in terms of **a** and **b**
24(b)	$\frac{5}{4}\mathbf{b}$	**3**	**M2** for $\overrightarrow{RS} = \frac{1}{4}\mathbf{b}$ oe or $\overrightarrow{MS} = \frac{3}{2}\left(\frac{1}{2}\mathbf{b}-\frac{2}{3}\boldsymbol{a}\right)$ oe or $\overrightarrow{NS} = \frac{1}{2}\left(\frac{1}{2}\mathbf{b}-\frac{2}{3}\boldsymbol{a}\right)$ oe or **M1** for a correct route in terms of vertices and/or **a** and/or **b** or **B**1 for answer $j\mathbf{b}$ where $j > 1$ or $\overrightarrow{RS} = \frac{1}{2}\overrightarrow{MQ}$, $\overrightarrow{RS} = \frac{1}{4}\overrightarrow{OR}$, oe $\overrightarrow{NS} = \frac{1}{2}\overrightarrow{MN}$, $\overrightarrow{MS} = \frac{3}{2}\overrightarrow{MN}$ $\overrightarrow{NS} = \frac{1}{3}\overrightarrow{MS}$

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